Section 2.1. Vector & Linear Equations

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A **linear equation** is an algebraic equation where each term has an exponent of 1 and when this equation is graphed, it always results in a **straight line**. This is the reason why it is named as a 'linear' equation.

Equations	Linear or Non-Linear
$y = 8x - 9$	Linear
$y = x^2 - 7$	Non-Linear, the power of the variable x is 2
$\sqrt{y} + x = 6$	Non-Linear, the power of the variable y is 1/2
$y + 3x - 1 = 0$	Linear
$y^2 - x = 9$	Non-Linear, the power of the variable y is 2

Two equations
Two unknowns

$$\begin{aligned} x - 2y &= 1 \\ 3x + 2y &= 11 \end{aligned}$$

Linear combination

$$3\begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

The left side of the vector equation is a **linear combination** of the column vectors. Our goal is to find the right numbers for **x = 3** and **y = 1**.This means we are using **multiplication and addition of vectors** together in one step.

We combine the equations into a matrix problem **Ax = b**:

Matrix equation

$$Ax = b$$

$$\begin{bmatrix} 1 & -2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Coefficient matrix A

The matrix form of the equations

We can multiply by rows or by columns. Either way, **Ax = b** must be a correct statement of the three equations.

Ax comes from **dot products**, each row times the column x:

$$Ax = \begin{bmatrix} (\textit{row 1}) \cdot x \\ (\textit{row 2}) \cdot x \\ (\textit{row 3}) \cdot x \end{bmatrix}.$$

Ax is a ***combination of column vectors***:

$$Ax = x\,(\textit{column 1}) + y\,(\textit{column 2}) + z\,(\textit{column 3})$$

Assignment for Section 2.1: Vectors and linear equations

(1) Compute

$$A\boldsymbol{x} = \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

in two ways:

(a) by dot products of the rows of A with the column vector $\boldsymbol{x}$;

(b) as a combination of the columns of A.

(2) Find the components of

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}.$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$ always yields the multiplication $Ix = x$

An **identity matrix** is a special square matrix where:

- All the numbers on the **main diagonal** (from top left to bottom right) are **1**,
- and **all other numbers are 0**.

11 Find the two components of Ax by rows or by columns:

$$\begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}.$$

17 Find the matrix P that multiplies (x, y, z) to give (y, z, x). Find the matrix Q that multiplies (y, z, x) to bring back (x, y, z).

18 What 2 by 2 matrix E subtracts the first component from the second component? What 3 by 3 matrix does the same?

$$E \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \qquad \text{and} \qquad E \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix}.$$

Section 2.2. The idea of Elimination

$$2x_1 - 3x_2 - 2x_3 + 2x_4 = 5$$

$$4x_1 - 2x_2 + 2x_3 + 2x_4 = 22$$

$$2x_1 - 3x_2 + 2x_3 + 10x_4 = 13$$

$$6x_1 - 9x_2 - 6x_3 + 6x_4 = 27$$

In matrix form:

$$\begin{bmatrix} 2 & -3 & -2 & 2 \\ 4 & -2 & 2 & 2 \\ 2 & -3 & 2 & 10 \\ 6 & -9 & -6 & 6 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 22 \\ 13 \\ 27 \end{pmatrix}$$

In matrix form:

$$\begin{bmatrix} 2 & -3 & -2 & 2 \\ 4 & -2 & 2 & 2 \\ 2 & -3 & 2 & 10 \\ 6 & -9 & -6 & 6 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 22 \\ 13 \\ 27 \end{pmatrix}$$

$$2x_1 - 3x_2 - 2x_3 + 2x_4 = 5$$

$$2x_2 + 3x_3 - x_4 = 6$$

$$4x_3 + 8x_4 = 8$$

$$2x_4 = 4$$

Upper Triangular System

$$x_1 - 2x_2 = 1$$

$$3x_1 + 2x_2 = 11$$

To eliminate x1 from the second row:

We need to subtract a multiple of equation 1 from equation 2.

$$x_1 - 2x_2 = 1$$

$$8x_2 = 8$$

The multiplier is 3.

Elimination produces an upper triangular system.

An **upper triangular system** is a set of equations where:

- Each equation has **fewer variables** than the one above it.
- The **coefficients below the main diagonal are all zero** (in matrix form).

$$\mathsf{U} = \begin{bmatrix} 0 & a_{12} & \cdots & a_{1\,n} \\ 0 & 0 & \cdots & a_{2\,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}.$$

1

$$4x_1 - 8x_2 = 4$$

$$3x_1 + 2x_2 = 11$$

The multiplier is 3/4.

2

$$x_1 - 2x_2 = 1$$

$$3x_1 - 6x_2 = 11$$

$$x_1 - 2x_2 = 1$$

$$0x_2 = 8$$

No solution

3

$$x_1 - 2x_2 = 1$$

$$3x_1 - 6x_2 = 3$$

$$x_1 - 2x_2 = 1$$

$$0x_2 = 0$$

Infinite many solutions

$$0x_1 + 2x_2 = 4$$

$$3x_1 - 2x_2 = 5$$

$$3x_1 - 2x_2 = 5$$

$$2x_2 = 4$$

Back substitution is a method used to solve an **upper triangular system** of equations.You start from the **bottom equation** (which has only one variable), find its value, and then **substitute it upward** to find the remaining variables.

Example:

Solve this system:

$$\begin{cases} x + 2y + 3z = 14 \\ y + 2z = 8 \\ z = 3 \end{cases}$$

Step 1: Start with the bottom equation

$$z = 3$$

Step 2: Substitute $z = 3$ into the second equation

$$y + 2(3) = 8 \Rightarrow y = 2$$

Step 3: Substitute $y = 2$ and $z = 3$ into the first equation

$$x + 2(2) + 3(3) = 14 \Rightarrow x + 4 + 9 = 14 \Rightarrow x = 1$$

x=1, y=2, z=3

Assignment for Section 2.2: The ideal of Elimination

(1) Consider the 2 by 2 system

$$\begin{aligned} 2x &- 4y = 6 \\ -x &+ 5y = 0 \end{aligned}$$

(a) What multiple of the first equation should be subtracted from the second equation?

(b) After this elimination step, solve the triangular system.

(c) If the right side changes to $(-6, 0)$, what is the new solution?

(2) Apply forward elimination and back substitution to solve

$$\begin{aligned} 2x &- 3y &&= 3 \\ 4x &- 5y + z &= 7 \\ 2x &- y - 3z &= 5 \end{aligned}$$

(3) Find the pivots and the solutions for the system

$$\begin{aligned} 2x &+ y &&&= 0 \\ x &+ 2y + z &&= 0 \\ &y + 2z + t &= 0 \\ &z + 2t &= 5 \end{aligned}$$

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PROBLEM I

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 0 & 1 \\ 1 & 4 & 5 \end{bmatrix} \quad \text{And} \quad B = \begin{bmatrix} 3 & 2 \\ 1 & 0 \\ 0 & 4 \end{bmatrix}$$

1) Compute AB
2) Switching the rows and columns of B gives a matrix

$$C = \begin{bmatrix} 3 & 1 & 0 \\ 2 & 0 & 4 \end{bmatrix}$$

Compute CAB

PROBLEM II

The linear combinations of $v = (1, 2, 3)$ and $w = (3, -2, 1)$ fill a plane in the x, y, z plane.

1)Find the components of $cv + dw$ with the number c and d.

2)Verify the zero vector $(0, 0, 0)$ is on the plane of v and w.

3)Verify that $n = (1, 1, -1)$ is perpendicular to the plane and find the unit vector in the same direction as n.

4)Use an equation with three unknowns to describe the plane and give the reason.

PROBLEM III

$$Let(A) = \begin{bmatrix} 1 & 3 & 1 \\ -1 & -2 & 1 \\ 3 & 7 & 5 \end{bmatrix}$$

1) For $B = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$, convert the system $Ax = b$ to an upper system $Ux = c$. Then find the solution by back substitution.

2) Find the inverse matrix A^{-1}

3) With the upper triangular U obtained in (1), find lower triangular L such that $A = LU$

Google : https://github.com/elalamiimed/HITSZCS And Find more about us!

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① $A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 0 & 1 \\ 1 & 4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 2 \\ 1 & 0 \\ 0 & 4 \end{bmatrix}$

$C = \begin{bmatrix} 3 & 1 & 0 \\ 2 & 0 & 4 \end{bmatrix}$

1) $\vec{AB} = \begin{bmatrix} 7 & 4 \\ 9 & 10 \\ 7 & 22 \end{bmatrix}$

2) $\vec{CAB} = \begin{bmatrix} 30 & 22 \\ 42 & 96 \end{bmatrix}$

② $\vec{v} = (1, 2, 3)$, $\vec{w} = (3, -2, 1)$

1) $c + 3d$, $2c - 2d$, $3c - d$.

2) $(0, 0, 0)$

Assume $c = 0$, $d = 0$

$0v + 0w = (0, 0, 0)$

The origin lies in the plane.

3) $n = (1, 1, -1)$

$\vec{n} \cdot \vec{v} = 1 \cdot 1 + 1 \cdot 2 + 3(-1) = 0$

$\vec{n} \cdot \vec{w} = 1 \cdot 3 - 2 \cdot 1 - 1 \cdot 1 = 0$

} Both are zero

$\vec{n} \perp \vec{v}$

$\vec{n} \perp \vec{w}$

$u_n = \frac{n}{\|n\|} = \frac{(1, 1, -1)}{\sqrt{1^2 + 1^2 + (-1)^2}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right)$.

4) Equation of the plane (three unknowns)

Since the plane passes through the origin and has normal $n = (1, 1, -1)$, its equation is

$n \cdot (x, y, z) = 0 \Rightarrow x + y - z = 0$

y can be $\pm$, but x and z are only in $+$.

As $n = (1, 1, -1)$ $\perp$ is in the plane.

③ $A = \begin{bmatrix} 1 & 3 & 1 \\ -1 & -2 & 1 \\ 3 & 7 & 5 \end{bmatrix}$

1) $B = \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$ $\left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ -1 & -2 & 1 & 4 \\ 3 & 7 & 5 & -1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ 0 & 1 & 2 & 5 \\ 0 & -2 & 2 & -4 \end{array}\right]$

$f_{21} = -1$

$f = 3$

$f_{32} = -2 \Rightarrow \left[\begin{array}{ccc|c} 1 & 3 & 1 & 1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 6 & 6 \end{array}\right]$

$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} -9 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$

$6x_3 = 6$ → $x_3 = 1$

$x_2 + 2x_3 = 5$ → $x_2 = 3$

$x_1 = 1 - 1 - 9$ → $x_1 = -9$

2) A^{-1}

$\begin{bmatrix} 1 & 3 & 1 & 1 & 0 & 0 \\ -1 & -2 & 1 & 0 & 1 & 0 \\ 3 & 7 & 5 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & -2 & 2 & -3 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & 0 & 6 & -1 & 2 & 1 \end{bmatrix}$

$f_{23} = \frac{2}{6} = \frac{1}{3}$, $f_{13} = \frac{1}{6}$

$\begin{bmatrix} 1 & 3 & 0 & \frac{7}{6} & -\frac{1}{3} & -\frac{1}{6} \\ 0 & 1 & 0 & 4/3 & 1/3 & -1/3 \\ 0 & 0 & 6 & -1 & 2 & 1 \end{bmatrix} \longrightarrow f_{12} = 3 \begin{bmatrix} 1 & 0 & 0 & -17/6 & -4/3 & 5/6 \\ 0 & 1 & 0 & 4/3 & 1/3 & -1/3 \\ 0 & 0 & 6 & -1 & 2 & 1 \end{bmatrix} \rightarrow$

$\frac{7}{6} - 4 = -\frac{17}{6}$

$-\frac{1}{3} - 1 = -\frac{4}{3}$

$-\frac{1}{6} - 3\left(-\frac{1}{3}\right) = -\frac{1}{6} + 1 = \frac{5}{6}$

$\rightarrow \begin{bmatrix} 1 & 0 & 0 & -17/6 & -4/3 & 5/6 \\ 0 & 1 & 0 & 4/3 & 1/3 & -1/3 \\ 0 & 0 & 1 & -1/6 & 1/3 & 1/6 \end{bmatrix}$

$A^{-1} = \frac{1}{6} \begin{bmatrix} -17 & -8 & 5 \\ 8 & 2 & -2 \\ -1 & 2 & 1 \end{bmatrix}$

3) $A = LU$

$L = \begin{bmatrix} 1 & 0 & 0 \\ +f_{21} & 1 & 0 \\ +f_{31} & +f_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ +1 & 1 & 0 \\ 3 & -2 & 0 \end{bmatrix}$

Thank you!
Good Luck on Mid-term<3
Special thanks to HSIYC Academic Department



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